HCI · MIDTERM PRESENTATION

Design and Evaluation of Haptic-assisted Motorcycling

GROUP 3

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Outline

01 Motivation

02 Related Works

03 Possible Designs

04 Experiment Framework

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06 Summary & Research Questions

SECTION 01

Motivation

The “Scooter Waterfall” at Taipei Bridge



Source: TVBS 地球黃金線 · 2023/07/17 · cars.tvbs.com.tw/car-news/129840

Term Definitions

大型重型機車

Big-sized Heavy Motorcycle

Engine ≥ 250 cc

Electric motor ≥ 40 HP

普通重型機車

Ordinary Heavy Motorcycle

Engine: 50–250 cc

Electric motor: 5–40 HP

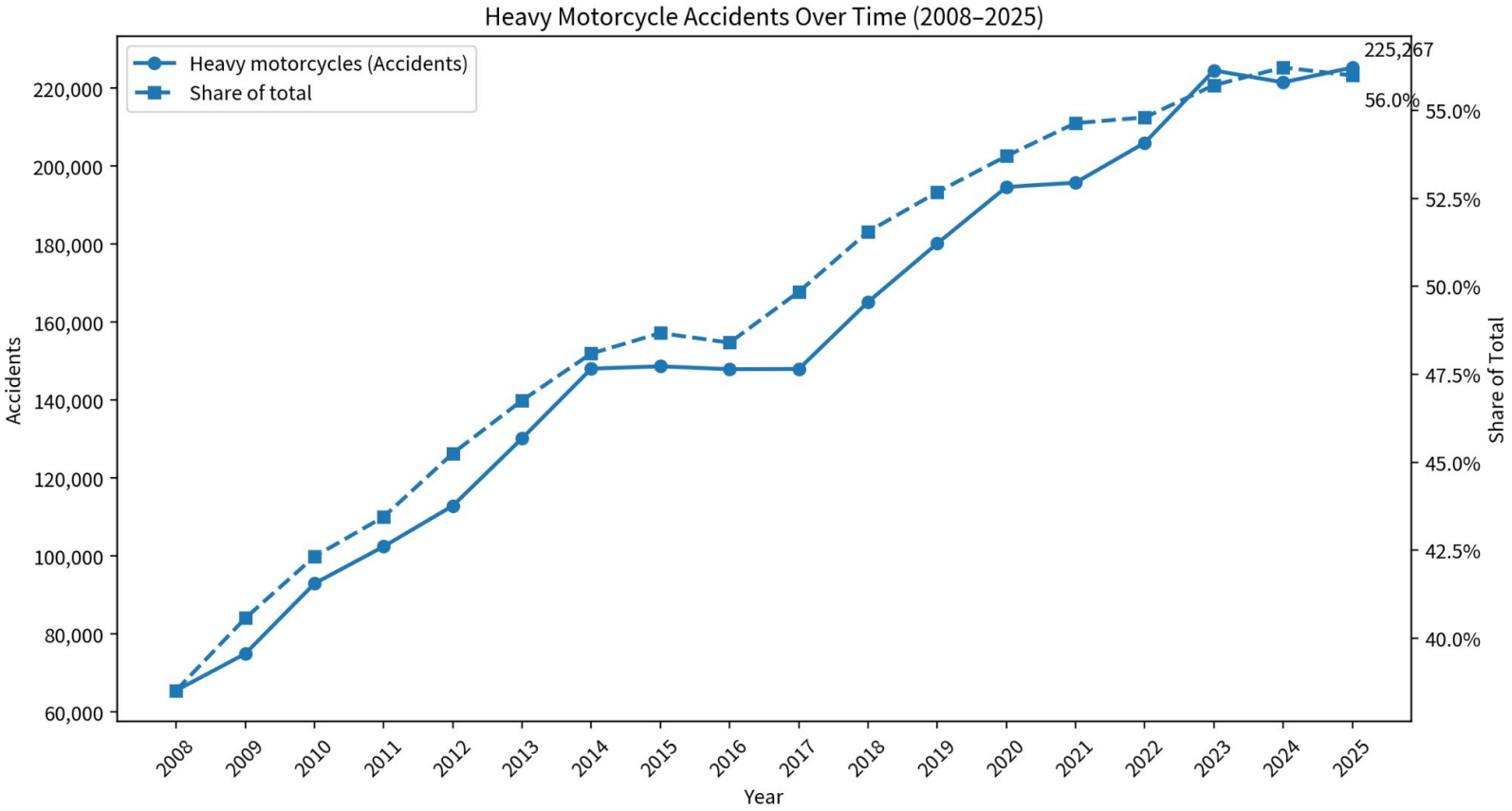
普通輕型機車

Ordinary Light Motorcycle

Engine ≤ 50 cc

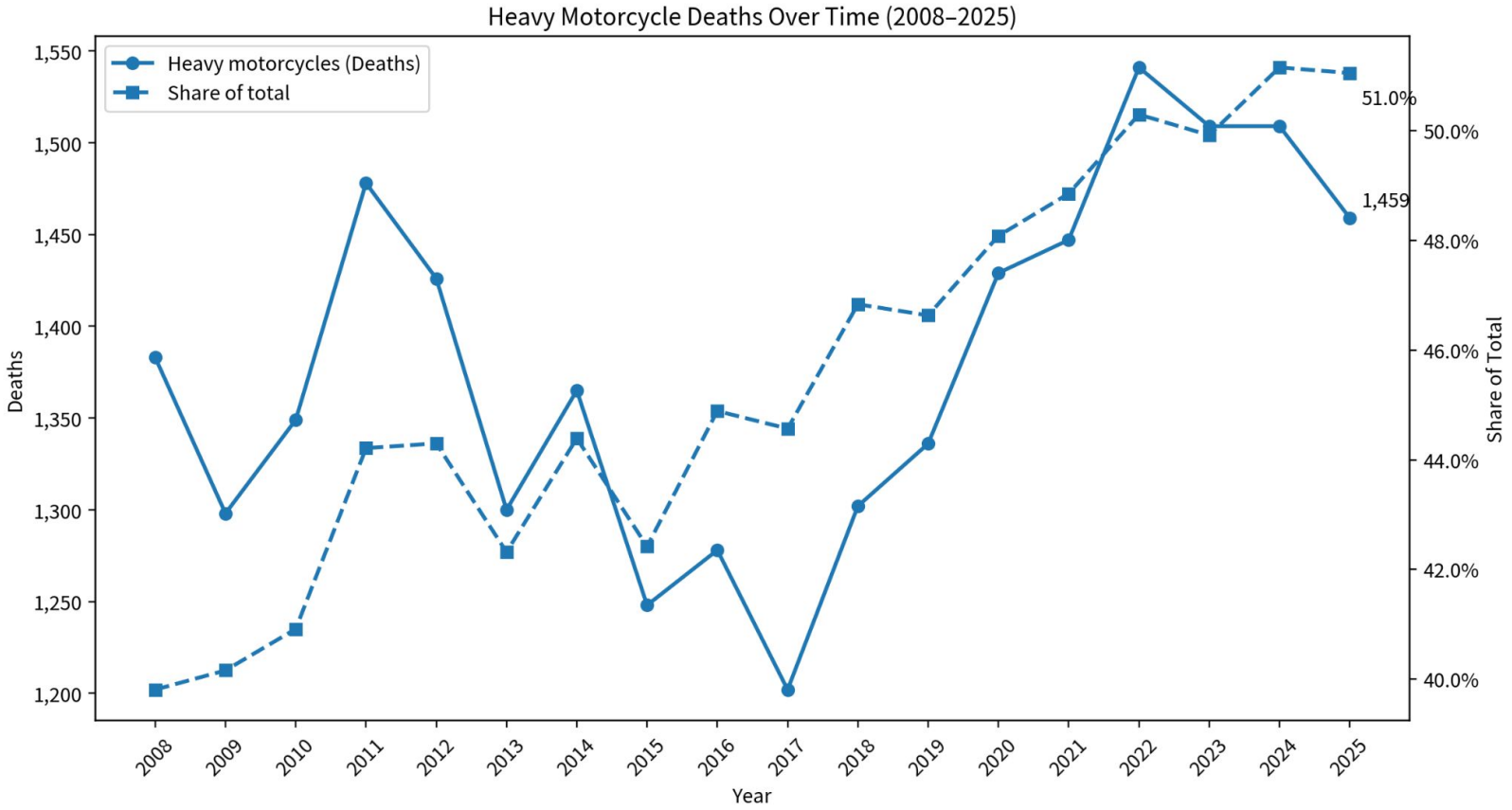
Electric: 1.34–5 HP; max speed > 45 km/h

Accidents over Time · 2008–2025



Source: Ministry of Transportation and Communications · statis.motc.gov.tw

Traffic Deaths over Time · 2008–2025



Source: Ministry of Transportation and Communications · statis.motc.gov.tw

Keep Your Eyes on the Road



SECTION 02

Related Works

Haptic Warning for Two-Wheelers

Prior studies explore how to deliver haptic information to riders.

Pressure-based handle

Three moving elements press against the rider's palm.

Perception & placement

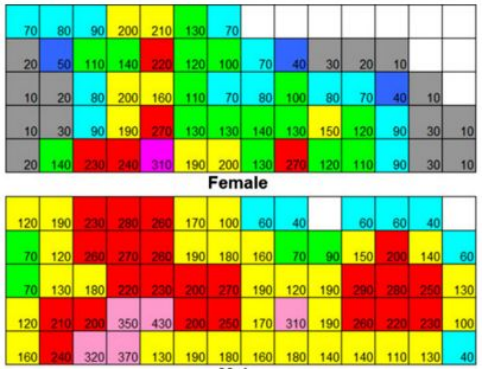
Contact-pressure measurements inform actuator placement.



(a) The vest. (b) The tactors. (c) The schematic.

Figure 1: The HaptiMoto vest with Lilypad Arduino (A), battery (B), BlueSMiRF (BT), and Vibe Boards (L,R,S).

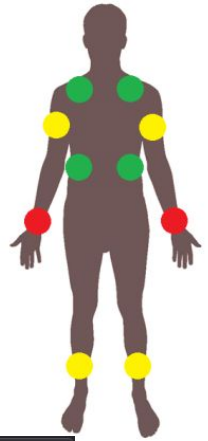




Wearable Haptic Feedback for Motorcyclists





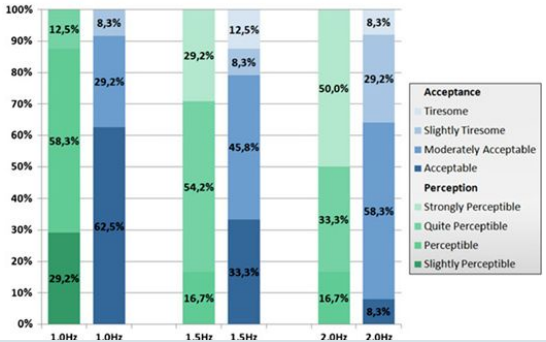


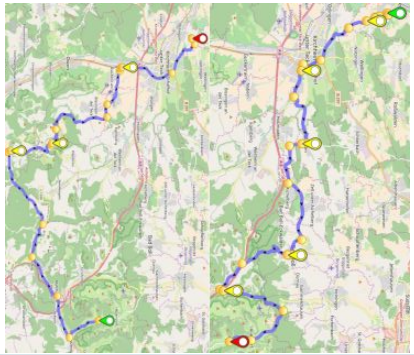
Rank	body location	avg Jerk
1	waist	1.86 m/s³
2	shoulders	1.92 m/s³
3	calf	2.84 m/s³
4	arm	3.50 m/s³
5	wrist	4.81 m/s³

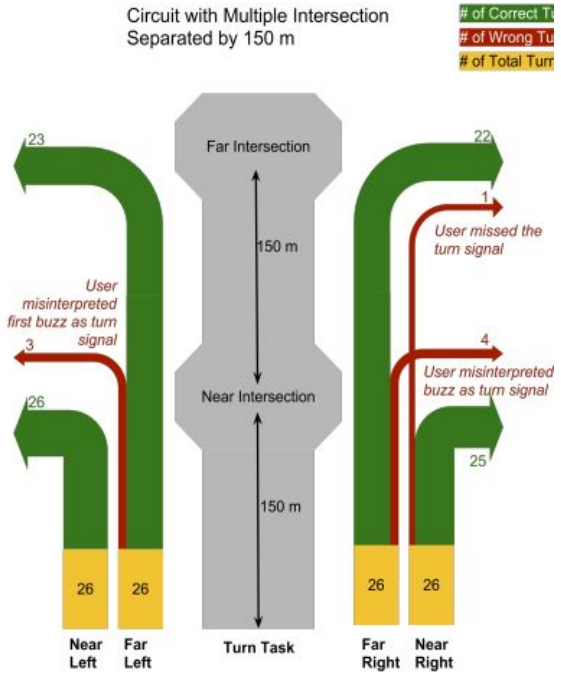
Gaps in the Experiments

Device feasibility has been studied; the next step is measuring how feedback affects riding tasks.

Early studies	Perceptibility and acceptability of haptic cues.
Navigation studies	Specific routes, turns and task-dependent error metrics.
Research opportunity	More comprehensive, end-to-end performance evaluation.







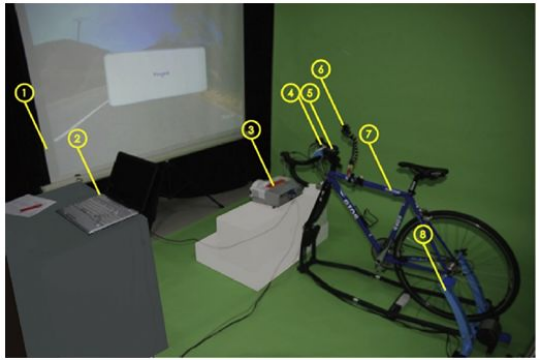
Motorcycle & Bicycle Simulators

Existing setups span multi-monitor rigs, physical controls and VR headsets.



Multi-monitor rig

Physical motorcycle controls



Bicycle simulator

Controlled hazard-perception tasks



VR simulator

Immersive head-tracked interaction

SECTION 03

Possible Designs

Handlebar-based Haptic Interface



Handlebar Vibration Prototype



Simulator — VR or Monitor-based?

Criterion	VR HMD	Single Monitor	Three Monitors
Turn navigation	Very suitable	Suitable	Suitable
Obstacle ahead	Very suitable	Very suitable	Very suitable
Blind spots	Free head turning	Limited field of view	Wide view
Handlebar operation	Requires practice	Directly visible	Directly visible
Motion sickness	Needs mitigation	Still possible	Still possible
Cost	Low if owned	Low	Higher

Current choice: VR — natural head movement for mirrors and blind-spot checks.

Checklist for VR Design

MUST HAVE

VR HMD + Unity PC	Use available equipment
Fixed handlebars + mount	DIY with existing bicycle / scooter parts
Steering / throttle / brake	Simple sensors and controllers
Left / right vibration actuators	Low-cost prototype
Virtual mirrors/map/sounds + head tracking	Implemented in Unity

OPTIONAL

Eye tracking	If supported by the headset

SECTION 04

Experiment Framework

Candidate Short Tasks

01 Turn Navigation

RIGHT TURN



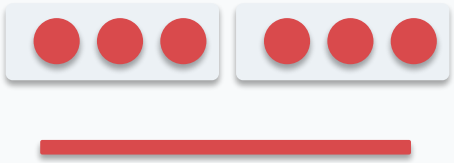
LEFT TURN



Wrong turns · Turn latency

02 Obstacle Ahead

SIMULTANEOUS



Sustained vibration

Brake RT · Collisions · TTC

03 Lane Change

LEFT-SIDE WARNING



RIGHT-SIDE WARNING



Frequently Vibrate

Unsafe changes · Safe gaps

Vibrates Inactive

Side obstacle: optional / pattern TBD

Experiment Conditions

	Single-context	Mixed-context
Without haptics	One selected task	Selected tasks in a random order
With haptics	Same task + haptic cue	Same tasks + matching haptic cues

Progressive evaluation Single task → Random non-overlapping tasks → Potentially overlapping tasks

SECTION 05

Limitations

Limitations & Future Work

01

Simulator vs. Real-world Riding

02

Real-world Handlebar Vibration

SECTION 06

Summary

RQ1

Can a convenient, non-wearable haptic feedback still outperform visual feedback in individual motorcycle riding tasks?

RQ2

How does performance change as task complexity increases, and can still outperform visual feedback?

RQ3

How the metrics collected in a low-cost motorcycle simulator reflect real-world riding performance, and what factors may limit their applicability?

Thank you.

Group 3 · Sheng-Yuan Huang · Yu-Chen Sun